Lecture 5

Van Kampen, surfaces and braids

Objectives. *We compute loop classes in a space assembled from open pieces. Van Kampen's theorem records how loops in the overlap become relations. For a surface presented by a polygon, its boundary gives the relation. For identical particles, the space consists of unordered configurations of distinct points. Loops in this space describe exchanges. Planar exchanges retain more information than their endpoint permutations, allowing arbitrary scalar phases.*

By the end of this lecture you will be able to:

- Apply van Kampen with path-connected open sets and intersection.
- Read the fundamental-group presentation of a polygonal surface.
- Relate planar exchanges to braids and three-dimensional exchanges to permutations.
- Derive the allowed one-dimensional exchange phases in each case.

5.1 Free groups and relations

A *word* in letters $a_1, \dots, a_k$ and their formal inverses is a finite string of these symbols. Delete adjacent inverse pairs until none remain. The resulting *reduced word* is unique. One algorithm reads from left to right, keeping a stack: cancel the last stored letter when the incoming letter is its inverse, and append it otherwise. Inserting an adjacent inverse pair leaves the final stack unchanged, even when the first inserted letter cancels a stored letter. The second then restores it. Thus reduction is independent of the order of deletions. A reduced word passes through the stack unchanged. Hence every complete sequence of deletions ends at the same stack output.

The *free group* F_k consists of reduced words. Multiply by concatenating and reducing. The empty word is the identity. Reverse a word and invert each letter to get its inverse. Reduction is unchanged by inserting or deleting inverse pairs in any context, so reducing after either stage of a triple concatenation gives the same result. This proves associativity. For example,

$$ab^{-1}ba^{-1}b = b, \qquad aba^{-1}b^{-1} \neq 1 \quad \text{in } F_2.$$

The second word is already reduced. A choice of images for the generators in any group extends uniquely to a homomorphism, by substituting in words. Inverse cancellations do not change the substituted group element.

A subgroup N of G is *normal* if $gNg^{-1} = N$ for every $g \in G$. The quotient G/N identifies elements differing by an element of N. Its multiplication is inherited from G; normality makes this independent of representatives. The *normal closure* of specified words is the smallest normal subgroup containing them, equivalently the subgroup generated by all their conjugates and inverses. A *presentation* is written

$$\langle a_1, \dots, a_k \mid r_1, \dots, r_s \rangle = F_k / \langle\langle r_1, \dots, r_s \rangle\rangle.$$

It imposes $r_j = 1$ in every word context, not merely deletion of one printed copy of a relator. For example, $\langle a \mid a^m \rangle \cong \mathbb{Z}_m$ for $m \geq 1$, where $\mathbb{Z}_m$ is the group of integers modulo m. The *free product* $G * H$ joins two groups without additional relations: words alternate nonidentity elements of the two groups, combining adjacent elements from the same group. In particular, $\mathbb{Z} * \mathbb{Z} = F_2$.

5.2 Van Kampen's theorem

Let $X = U \cup V$, with $x_0 \in U \cap V$. The inclusions induce maps from the fundamental groups of these sets. Write i_{U*} and i_{V*} for the maps from the intersection to the two pieces.

Theorem 5.1 (Van Kampen · *cited; surjectivity sketched*)**.** *Suppose U and V are open and path-connected, and $U \cap V$ is path-connected. The natural homomorphism*

$$\pi_1(U, x_0) * \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0)$$

is surjective. Its kernel is the normal closure of $i_{U*}(w) i_{V*}(w)^{-1}$ for $w \in \pi_1(U \cap V, x_0)$.

This is Hatcher, Theorem 1.20. It identifies the two images of each overlap loop. We sketch the surjectivity construction. Subdivide a loop into pieces γ_j in U or V, using the Lebesgue-number lemma. Merge successive pieces in the same set. At a change of set the joining point is in $U \cap V$. Choose a path a_j in the intersection from x_0 to each joining point, with constant paths at the start and end. The based loops $a_{j-1} \cdot \gamma_j \cdot \bar{a}_j$ each lie in one piece. Their product represents the original loop, since the inserted reverse paths cancel. The kernel statement is cited, not proved by this construction.

The connected-intersection hypothesis matters. On S^1, take the north point $N = (0, 1)$ and south point $S = (0, -1)$. Cover the circle by their complements. Both sets are open intervals and have trivial fundamental group. Their intersection has two components. The circle group is $\mathbb{Z}$, so the conclusion of the theorem cannot hold without the hypothesis.

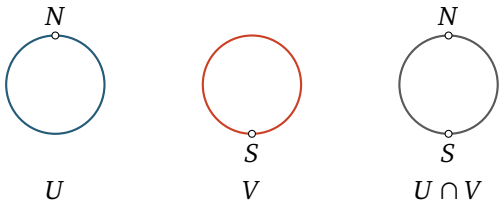


Figure 5.1: Removing N or S gives an open arc. Removing both leaves two components, so this circle cover does not satisfy van Kampen's hypotheses.

The *wedge* $S^1 \vee S^1$ identifies one point of each of two circles to a common point x_0. Remove a non-base point y_b of the second circle to get U, and a non-base point y_a of the first to get V. These are open in the wedge. The first retracts onto the first circle, the second onto the second circle. Collapse the remaining open arcs linearly to x_0; these homotopies fix their retained circles. The formulas agree at x_0, so finite closed pasting proves continuity. The intersection is four open arms joined at x_0 and contracts there. Hence the cited theorem gives

$$\pi_1(S^1 \vee S^1, x_0) \cong \mathbb{Z} * \mathbb{Z} = F_2.$$

The loops a and b run round the two circles. Inductively the same open-neighbourhood construction gives F_r for a finite wedge of r circles.

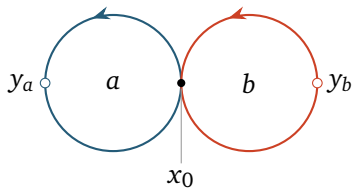


Figure 5.2: Removing the marked point from the opposite circle leaves an open neighbourhood that retracts onto the chosen circle. Both circles share x_0.

5.3 Polygonal surfaces

A *surface* is a Hausdorff space with a countable basis in which every point has a neighbourhood homeomorphic to an open disc in $\mathbb{R}^2$. A compact surface with these local models is called *closed*. For $g \geq 1$, let Σ_g be the standard closed surface obtained from a $4g$-gon by pairing equally labelled edges in the boundary word

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_g b_g a_g^{-1} b_g^{-1}.$$

The integer g is its *genus*. The paired arrows specify the same direction along each identified edge. Edge interiors join two half discs, and the cyclic corner identifications join to a disc. All vertices become one vertex, and the edges become a wedge of $2g$ circles. This is the polygonal model used in Hatcher, §1.2.

Proposition 5.2 (Surface groups · *proved given van Kampen*). *For $g \geq 1$, the fundamental group is*

$$\pi_1(\Sigma_g) \cong \left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{j=1}^{g} [a_j, b_j] \right\rangle, \qquad [a, b] = aba^{-1}b^{-1}.$$

Proof. Choose a point o in the face interior. Let $U = \Sigma_g \setminus \{o\}$ and let V be a small open disc about o inside that face. The punctured polygon retracts radially to its boundary, fixing it. This deformation respects the edge identifications and retracts U onto the wedge of $2g$ circles. The set V is contractible. The overlap is a punctured disc, with fundamental group $\mathbb{Z}$. Its positive generator follows the face boundary once after radial retraction. Choose a path from its base point to the polygon vertex to read the word in the stated generators. The result is $[a_1, b_1] \cdots [a_g, b_g]$. Van Kampen kills exactly its normal closure, giving the presentation. A different connecting path conjugates the relator and leaves its normal closure unchanged. ∎

For $g = 1$ the surface is the torus of Lecture 2; we compute its presentation explicitly in the final example.

5.4 Configuration spaces and endpoint permutations

Fix integers $n \geq 2$ and $d \geq 2$. The *ordered configuration space* and *unordered configuration space* are

$$\mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d) = \{(x_1, \ldots, x_n) \in (\mathbb{R}^d)^n : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Conf}_n(\mathbb{R}^d) = \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^d)/S_n.$$

We keep the dimension in the notation to distinguish the plane from space. In the planar case this is the space $C_n = \mathrm{Conf}_n(\mathbb{R}^2)$ of the endpoint-permutation convention. Here S_n is the group of permutations of $\{1, \dots, n\}$, with right-to-left composition. Its action is $(s \cdot x)_i = x_{s^{-1}(i)}$, which forgets labels in the quotient. The projection is a covering: take disjoint small balls around the n points. Their product and its $n!$ permuted copies are disjoint sheets over the same quotient neighbourhood. The quotient map is open because the saturation of an open set is a finite union of its permuted copies.

A loop downstairs lifts to n labelled particle trajectories. Label their initial positions $1, \dots, n$. The *outgoing permutation* $p_\gamma(i)$ is the final position of the particle starting at i. With the first-path-first convention,

$$p_{\gamma\cdot\delta} = p_\delta \circ p_\gamma, \qquad \rho([\gamma]) = p_\gamma^{-1}, \qquad \rho([\gamma][\delta]) = \rho([\gamma])\rho([\delta]).$$

Thus the inverse outgoing permutation is the homomorphism. Homotopy lifting makes it well-defined on loop classes. For the noncommuting test $p_\gamma = (12)$ and $p_\delta = (13)$, it gives

$$p_{\gamma\cdot\delta} = (123), \qquad \rho([\gamma][\delta]) = (132) = (12)(13).$$

In the plane, fix an unordered base configuration x_0. A *geometric braid* is n collision-free trajectories $(z_j(t), t)$ with that fixed set of positions at both $t = 0$ and $t = 1$. Time increases along every strand. Deformations keep the endpoint positions fixed and preserve the collision-free condition. Write B_n for their classes, with multiplication by running the first braid and then the second.

Proposition 5.3 (Exchange classes · *sketched; proved in* $\mathbb{R}^3$)**.** *In three dimensions,* $\pi_1(\mathrm{Conf}_n(\mathbb{R}^3)) \cong S_n$. *In the plane, the fundamental group is the braid group* B_n.

For $\mathbb{R}^3$, we prove that the ordered space is simply connected. Its collision sets are linear subspaces $L_{ik} = \{x_i = x_k\}$, for $1 \le i < k \le n$, of $\mathbb{R}^{3n}$, each of dimension $3n - 3$. A finite union of proper linear subspaces misses a point. Indeed, for each subspace choose a nonzero linear functional vanishing on it. On the curve $(1, t, \dots, t^{3n-1})$ each functional becomes a nonzero polynomial, with finitely many roots. Choose t outside those finite root sets.

First, a loop γ has positive distance ε from the closed union of collision sets, by compactness. Uniform continuity gives a subdivision $0 = t_0 < \cdots < t_m = 1$ with each image piece of diameter less than $\varepsilon/4$. Join successive vertices $v_j = \gamma(t_j)$ by straight segments to form η. At matching parameters, the interpolation between η and γ stays within $\varepsilon/2$ of $\gamma(t)$. This is a based homotopy avoiding collisions.

Next, choose an apex z_* outside all subspaces $L_{ik} + \mathrm{span}\{v_j, v_{j+1}\}$. Each has dimension at most $3n - 1$, so the finite-union argument applies. With two spatial coordinates, that sum can fill the ambient space, so the apex argument does not apply. The triangle with vertices v_j, v_{j+1}, z_* misses L_{ik}. Otherwise a collision point with nonzero apex coefficient would express z_* as an element of that forbidden subspace. A collision with zero apex coefficient would lie on the already collision-free polygon edge. Thus the map $h\colon [0,1]^2 \to \mathrm{Conf}_n^{\mathrm{ord}}(\mathbb{R}^3)$ given by

$$h(t, s) = (1 - s)\eta(t) + sz_*$$

contracts the loop freely. Lecture 3's moving-base-point construction turns this into a based contraction. Apply the same apex choice to two configurations instead of polygon vertices to join them by two straight segments. Hence the ordered space is path-connected and simply connected.

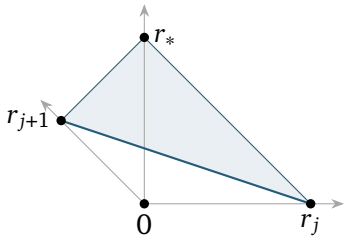


Figure 5.3: For two particles, project to relative coordinates $r = x_1 - x_2$. The projected vertices and apex span a triangle missing the collision point 0.

Coordinate permutations are homeomorphisms commuting with the projection, so they are deck transformations. They act transitively on a fibre and exhaust the deck group by uniqueness from one point. Now Corollary 4.4 gives S_n. The permutation reaching the lifted endpoint is $\rho([\gamma])$, by our chosen action, so this is the isomorphism stated above. Simon, §3.3.2, describes the exchange-space conclusion.

For the plane we sketch the identification. Lifting a configuration loop labels its strands. Conversely, their unordered positions give a configuration loop. The same correspondence applies to homotopies. Thus the two groups are identified. This is Simon, §3.3.1, in the course's path convention.

For two planar particles, write the positions as $c+u$ and $c-u$ in $\mathbb{C}$, where $u \neq 0$. Forgetting labels identifies u with $-u$. The centre c, radius $|u|$ and squared direction $(u/|u|)^2$ give

$$\operatorname{Conf}_2(\mathbb{R}^2) \cong \mathbb{R}^2 \times (0,\infty) \times S^1.$$

The inverse chooses either square root of the direction; the two choices give the same unordered pair. Locally a square-root branch is continuous, so the inverse is continuous. Contracting the first two factors gives $B_2 \cong \mathbb{Z}$. The counter-clockwise exchange $u(t) = e^{\pi \mathrm{i} t}$ makes the squared direction wind once. Two exchanges have winding two, even though their endpoint permutation is the identity.

5.5 Braid relations and exchange phases

Arrange the base positions on a real line. Let σ_i exchange positions i and $i+1$ counter-clockwise, fixing the other positions. Its inverse exchanges them clockwise. We cite Artin's presentation:

$$B_n = \left\langle \sigma_1, \dots, \sigma_{n-1} \;\middle|\; \begin{array}{ll} \sigma_i\sigma_j = \sigma_j\sigma_i, & |i-j| \geq 2, \\ \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}, & 1 \leq i \leq n-2 \end{array} \right\rangle.$$

The first relation exchanges separated pairs in either order. The second changes the order of three adjacent exchanges. The completeness of these relations is cited from Simon, §3.3.1 and Exercise 3.1. His braid words are read rightmost first and his positive exchange is clockwise. Our words are read leftmost first and positive exchanges are counter-clockwise. His positive generator is therefore our σ_i^{-1}.

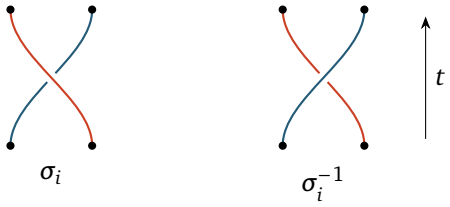


Figure 5.4: Time runs upward and the real axis to the right. At a crossing, the strand with larger imaginary coordinate is drawn on top.

A one-dimensional unitary representation is a homomorphism $\chi\colon G \to U(1)$, where $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. Its value on an exchange is the *exchange phase*.

> **Corollary 5.4** (Scalar exchange phases · *proved given the presentations*)**.** *For $n \geq 2$, the one-dimensional unitary representations of S_n give exchange phases $+1$ or -1. For B_n, any phase $e^{i\theta}$, with $\theta \in \mathbb{R}$, is possible, with the same phase on every σ_i.*

Proof. Adjacent transpositions generate S_n and are conjugate to one another. Scalar images of conjugate elements agree. Each transposition squares to the identity, so its common image q satisfies $q^2 = 1$. Both possibilities occur: the constant character and the sign character, the latter given by the determinant of the permutation matrix.

For B_n, write $q_i = \chi(\sigma_i)$. All q_i are nonzero scalars. The adjacent braid relation gives

$$q_i^2 q_{i+1} = q_i q_{i+1}^2, \qquad q_i = q_{i+1}.$$

For $n \geq 3$ this makes all images equal. For $n = 2$ there is only one generator. Conversely, assigning the same $q \in U(1)$ to every generator satisfies both families of defining relations. Thus every $q = e^{i\theta}$ defines a character. ∎

Thus a character with value $e^{i\theta}$ on our positive exchange has value $e^{-i\theta}$ on Simon's positive exchange. The choices $q = 1$ and $q = -1$ describe scalar bosonic and fermionic statistics. The other planar choices describe *abelian anyons*. Higher-dimensional unitary representations of S_n give matrix-valued exchange operators and the possibility of parastatistics. The scalar conclusion does not classify those representations.

5.6 The torus square

We compute the group of the square with opposite sides identified in the same direction. The quotient is the torus T^2 of Lecture 2. All four corners give one base point, and the two edge loops are a and b.

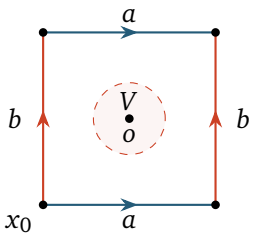


Figure 5.5: The paired arrows give $aba^{-1}b^{-1}$ on the counter-clockwise boundary. The dashed disc V and puncture o give the open cover imposing that relator.

Removing the centre retracts the square quotient onto the two edge loops, so its group is F_2. Adding the centre disc imposes the boundary relator by the open cover used in Proposition 5.2. Therefore

$$\pi_1(T^2) \cong \langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2.$$

For the last isomorphism, the relation makes a and b commute, so every word becomes $a^m b^n$. The exponent-sum map to $(m, n) \in \mathbb{Z}^2$ kills the relator and is inverse to $(m, n) \mapsto a^m b^n$. This agrees with the coordinate-winding calculation of Lecture 3.

Self-check. Prove that the group of $S^1 \vee S^1$ is nonabelian by finding a surjective homomorphism onto S_3.

Answer. Send a to (12) and b to (13). The universal property of F_2 gives a homomorphism. Its image contains $(23) = (12)(13)(12)$ and the two three-cycles, as well as the identity. Thus it contains all six permutations. Since $(12)(13) \neq (13)(12)$, the image is nonabelian, and F_2 is nonabelian. In particular, its commutator $aba^{-1}b^{-1}$ cannot be removed before a face imposes that relation.

Summary. Van Kampen combines path-connected open pieces with path-connected intersection by imposing the overlap relations. A polygonal surface has one relation, the product of its handle commutators. Planar configuration loops are braids; in three dimensions the ordered-space contraction reduces loop classes to permutations. The inverse outgoing permutation is the homomorphism in our path convention. Scalar exchange phases are arbitrary for braids and restricted to ± 1 for permutations. Lecture 6 counts holes of every dimension by linear algebra.

Reading. Hatcher, *Algebraic Topology*, §1.2; Nakahara, *Geometry, Topology and Physics*, §4.4; Simon, *Topological Quantum*, §§3.3–3.5 and Exercise 3.1.